# Lab#02
# Implementation of Scanner class for reading and taking inputs in java.

## Objective

To get familiarity with the Scanner Class for reading and taking inputs in java.

## Lab Goals

To demonstrate and gain familiarity with the Scanner class in Java for reading and taking inputs:

a. **Understand Scanner Class Basics:** Gain a foundational understanding of the Scanner class in Java, including its purpose, syntax, and basic usage for reading input from various sources such as the console, files, or strings.

b. **Practice Input Reading:** Learn how to use the Scanner class to read different types of input data, including integers, floating-point numbers, strings, and characters, and understand the methods provided by the Scanner class for retrieving this input.

## PRINT FUNCTION IN JAVA

**j**ava **System.out.println()** is used to print an argument that is passed to it. The statement can bebroken into 3 parts which can be understood separately as:

1. **System:** It is a final class defined in the java.lang package.
2. **out:** This is an instance of PrintStream type, which is a public and static member field Of the System class.
3. **println():** As all instances of PrintStream class have a public method println(), hence we can invoke the same on out as well. This is an upgraded version of print(). It prints any argument passed to it and adds a new line to the output. We can assume that System.out represents the Standard Output Stream.

Syntax:

4. System.out.println(*parameter*)

**Example**

```
class demo {
    public static void main(String[]args){
        System.out.println("Welcome");
        System.out.println("To");
        System.out.println("ssuet");
    }
}
```

## Exercise

A. Write a program to make BIO_DATA and take several user inputs (i.e. name, age, gender, department, email id, father name) and print the result along with the greeting message.

## Program:

```java
import java.util.Scanner;

public class Lab2 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter Name: ");
        String name = sc.nextLine();

        System.out.print(s: "Enter Age: ");
        int age = sc.nextInt();
        sc.nextLine();

        System.out.print(s: "Enter Gender: ");
        String gender = sc.nextLine();

        System.out.print(s: "Enter Department: ");
        String department = sc.nextLine();

        System.out.print(s: "Enter Email ID: ");
        String email = sc.nextLine();

        System.out.print(s: "Enter Father's Name: ");
        String fatherName = sc.nextLine();
```

```java
        System.out.println(x: "\n========== BIO DATA ==========");
        System.out.println("Hello, " + name + "! Here is your Bio
        System.out.println(x: "------------------------------");
        System.out.println("Name        : " + name);
        System.out.println("Age         : " + age);
        System.out.println("Gender      : " + gender);
        System.out.println("Department  : " + department);
        System.out.println("Email ID    : " + email);
        System.out.println("Father Name : " + fatherName);
        System.out.println(x: "==============================");
        sc.close();
    }
}
```

## Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 2> java Lab2
Enter Age: 19
Enter Gender: Female
Enter Department: BSET
Enter Email ID: ansharahkhan81@gmail.com
Enter Father's Name: Imran Javed

========== BIO DATA ==========
Hello, Ansharah! Here is your Bio Data:
------------------------------
Name        : Ansharah
Age         : 19
Gender      : Female
Department  : BSET
Email ID    : ansharahkhan81@gmail.com
Father Name : Imran Javed
==============================
PS C:\Users\Ansharah\Desktop\Java\Lab 2>
```

B. Write a program which takes two numbers as input from user and calculate sum and difference of both numbers.

**Program:**

```java
import java.util.Scanner;

public class Lab2 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print(s: "Enter second number: ");
        double num2 = sc.nextDouble();

        double sum        = num1 + num2;
        double difference = num1 - num2;

        System.out.println("\nFirst Number  : " + num1);
        System.out.println("Second Number : " + num2);
        System.out.println("Sum           : " + sum);
        System.out.println("Difference    : " + difference);
        sc.close();
    }
}
```

**Output:**

```
PS C:\Users\Ansharah\Desktop\Java\Lab 2> java Lab2
Enter first number: 45
Enter second number: 67

First Number  : 45.0
Second Number : 67.0
Sum           : 112.0
Difference    : -22.0
PS C:\Users\Ansharah\Desktop\Java\Lab 2>
```

C. Take a three-digit number from user, separate each digit of the number and sum them For

example: 123

1+2+3= 6

**Program:**

```java
import java.util.Scanner;

public class Lab2 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter a 3-digit number: ");
        int number = sc.nextInt();

        // Separate each digit
        int digit1 = number / 100;
        int digit2 = (number / 10) % 10;
        int digit3 = number % 10;

        int sum = digit1 + digit2 + digit3;

        System.out.println("\nNumber  : " + number);
        System.out.println("Digits  : " + digit1 + ", " + digit2 + ", " + digit3);
        System.out.println(digit1 + " + " + digit2 + " + " + digit3 + " = " + sum);
        sc.close();
    }
}
```

## Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 2> javac Lab2.java
PS C:\Users\Ansharah\Desktop\Java\Lab 2> java Lab2
Enter a 3-digit number: 567

Number  : 567
Digits  : 5, 6, 7
5 + 6 + 7 = 18
PS C:\Users\Ansharah\Desktop\Java\Lab 2>
```